

Static Electricity – Dancing Cereal



Can a Comb Make Cereal Dance?

Hook & Wonder: The Magic of Static Electricity

Can a comb really make cereal dance without touching it? Watch as puffed rice (murmura) magically jumps off a steel thali toward a simple plastic comb!



What is Static Electricity?

Static Electricity

Static electricity is an invisible force created when we rub two things together. When you rub a comb through your hair, tiny charges build up and create this magical pulling power!



3

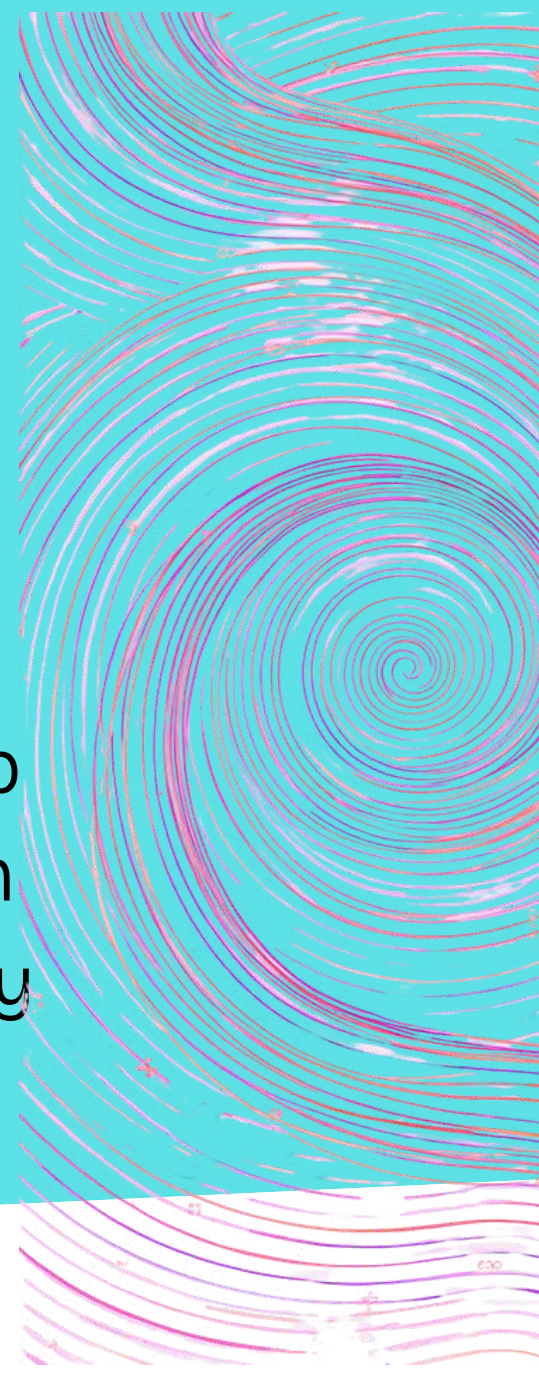




How Charge Builds Up

Rubbing causes tiny invisible charges to build up on objects.

When you rub a comb through your hair, tiny charges move from your hair to the comb. The comb gains negative charges (-) while your hair loses them and becomes positive (+). This is how static electricity is created!



Why Does Static Electricity Attract?

Attract • Force

When you rub a comb on your hair, it collects invisible charges. These charges create a pulling force that attracts light objects like murmura toward the comb — just like a magnet!



Indian Examples of Static Electricity

Science is all around us in daily life!

- Plastic “Comb” pulling hair every morning
- Crackling woollen sweater in Delhi winter
- Balloon sticking to wall during Diwali
- Shock from metal door handle after sitting on plastic chair
- Lightning — nature's giant static spark



Materials for Our Dancing Cereal Experiment

Gather these simple items to explore static electricity!

- Plastic comb
- Murmura (puffed rice)
- Steel thali
- Woollen cap
- Balloon
- Paper bits
- Salt grains
- Dry tea leaves
- Plastic scale



Investigation Steps: Predict → Charge → Test → Observe

**Follow the scientific method
to discover static electricity!**



1. Predict: Guess which objects will jump
2. Charge: Rub comb/balloon on hair or wool
3. Test: Hold charged comb near materials
4. Observe & Record: See what moves!

Control Test: Charged vs Uncharged Comb

See the difference charge makes!

Left: Uncharged comb – murmura stays still, no movement at all. Right: Charged comb – murmura pieces jump and dance toward the comb like magic!



Predict Which Objects Will Dance?

Make your guesses before we test!

- ☐ Murmura (puffed rice) — ?
- ☐ Paper bits — ?
- ☐ Stone or coin — ?
- ☐ Salt grains — ?
- ☐ Pencil — ?



Testing & Observing with Dancing Cereal

Hands-on experiment time!

Watch the magic happen as murmura jumps to the comb! Rub your comb on your hair, hold it near the puffed rice, and see the cereal dance without touching it!



What Did We See? Observations

Our experiment results show how static electricity works!

- Murmura and paper bits jump — very light!
- Stone and pencil don't move — too heavy
- Salt grains jump — super light!
- Static electricity only attracts light, uncharged objects



Discussion: What Did You Notice?

13

Let's share our observations and discoveries!

- Why did some things jump and others didn't?
- When do you see static electricity in your home?
- How does weather (dry vs humid) affect this?

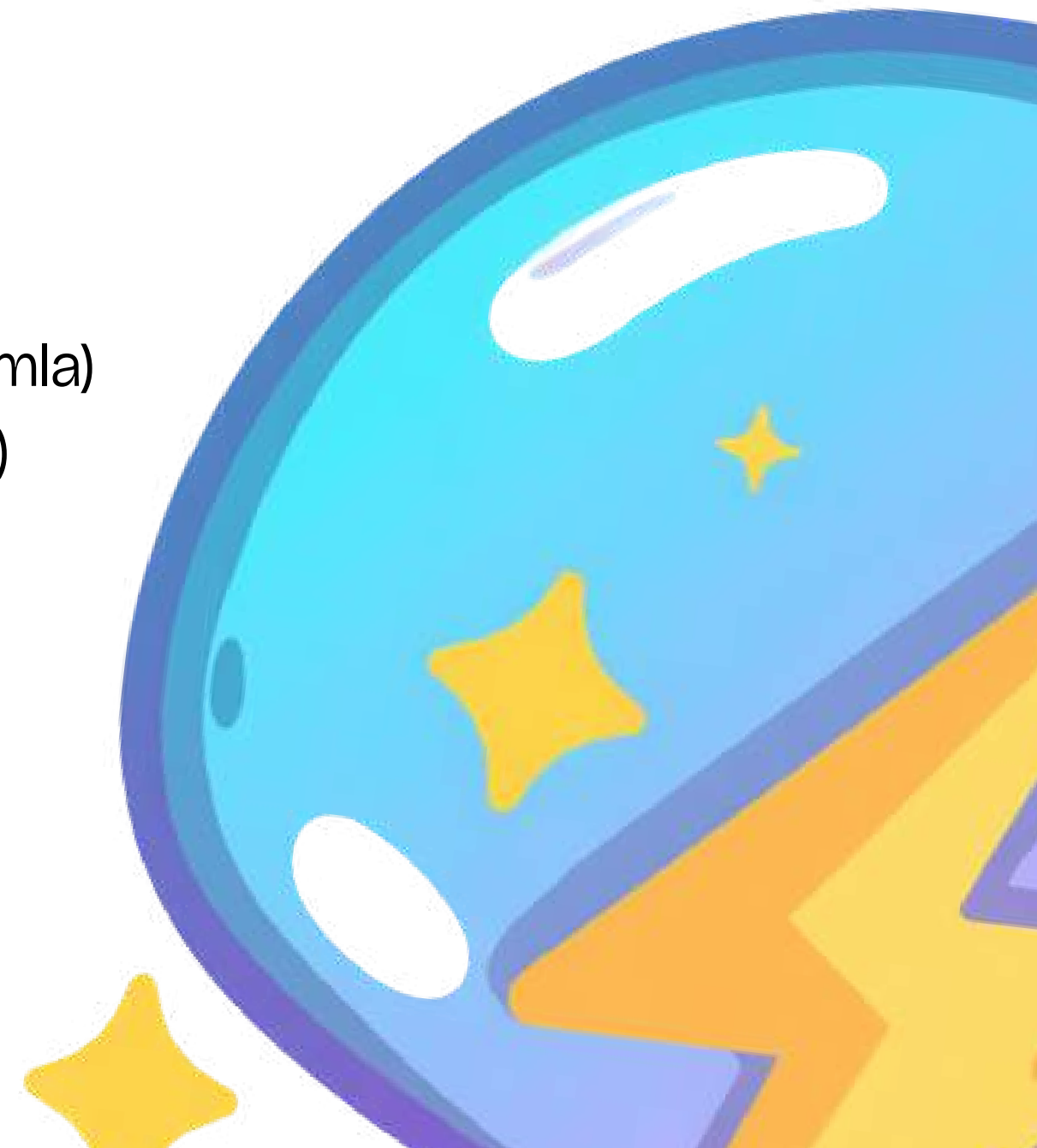




Real-Life Connections in India

Science is all around us!

- Comb pulling hair (morning routine)
- Crackling sweater in winter (Delhi, Shimla)
- Balloons at festivals (Diwali, birthdays)
- Lightning during monsoon
- Plastic wrapping sticking to lunchbox



3 Big Things We Learned Today

Let's remember the magic of static electricity!

1. Static electricity is made by rubbing things together
2. It creates an invisible force that pulls light objects like murmura
3. We can predict, test, and observe how this force works!



Exit Ticket: Static or Not Static?

Is this static electricity?
Yes or No?

- Comb pulling hair strands
- Shock from metal door handle
- Stone sitting on a table
- Balloon sticking to wall





Key Vocabulary

Important Terms with Hindi Transliteration

- ⚡ STATIC ELECTRICITY- Invisible force from rubbing
- ⚡ CHARGE - Energy that builds up
- ⚡ ATTRACT - Pull things closer
- ⚡ FORCE - Push or pull power
- ⚡ DISCHARGE - ZAP! Release of charge
- ⚡ LIGHTNING - Nature's giant spark

17



Take-Home Challenge

Keep exploring static electricity at home!

Try rubbing different things together — a balloon on your hair, a plastic scale on wool, or a comb on a sweater. See if you can make paper bits dance! Watch for static shocks and tell a family member about the science behind it.



Thank You & Keep Exploring!

Thank you for discovering the magic of static electricity with us! Keep experimenting and stay curious. Visit us at planrsacademy.com

